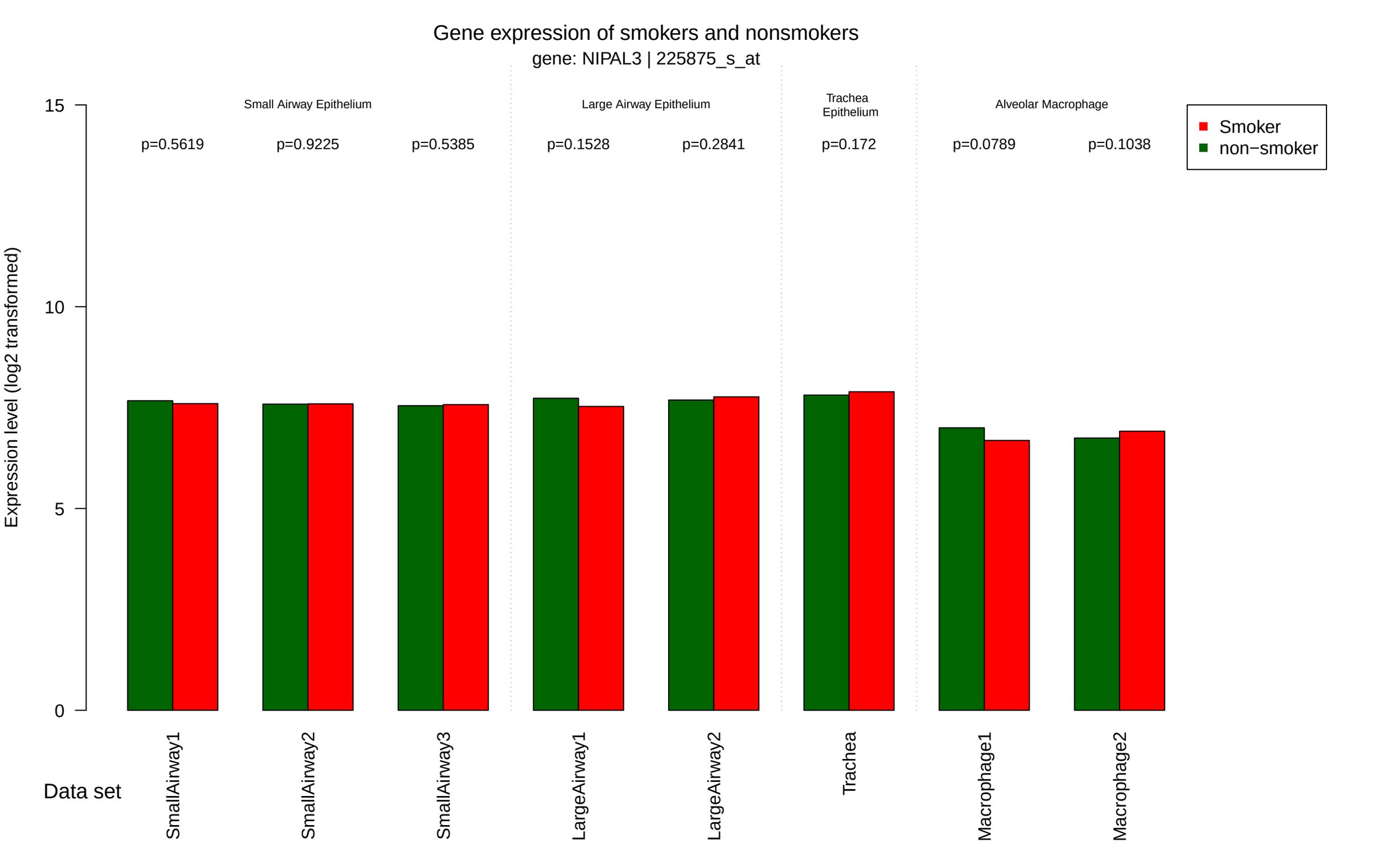
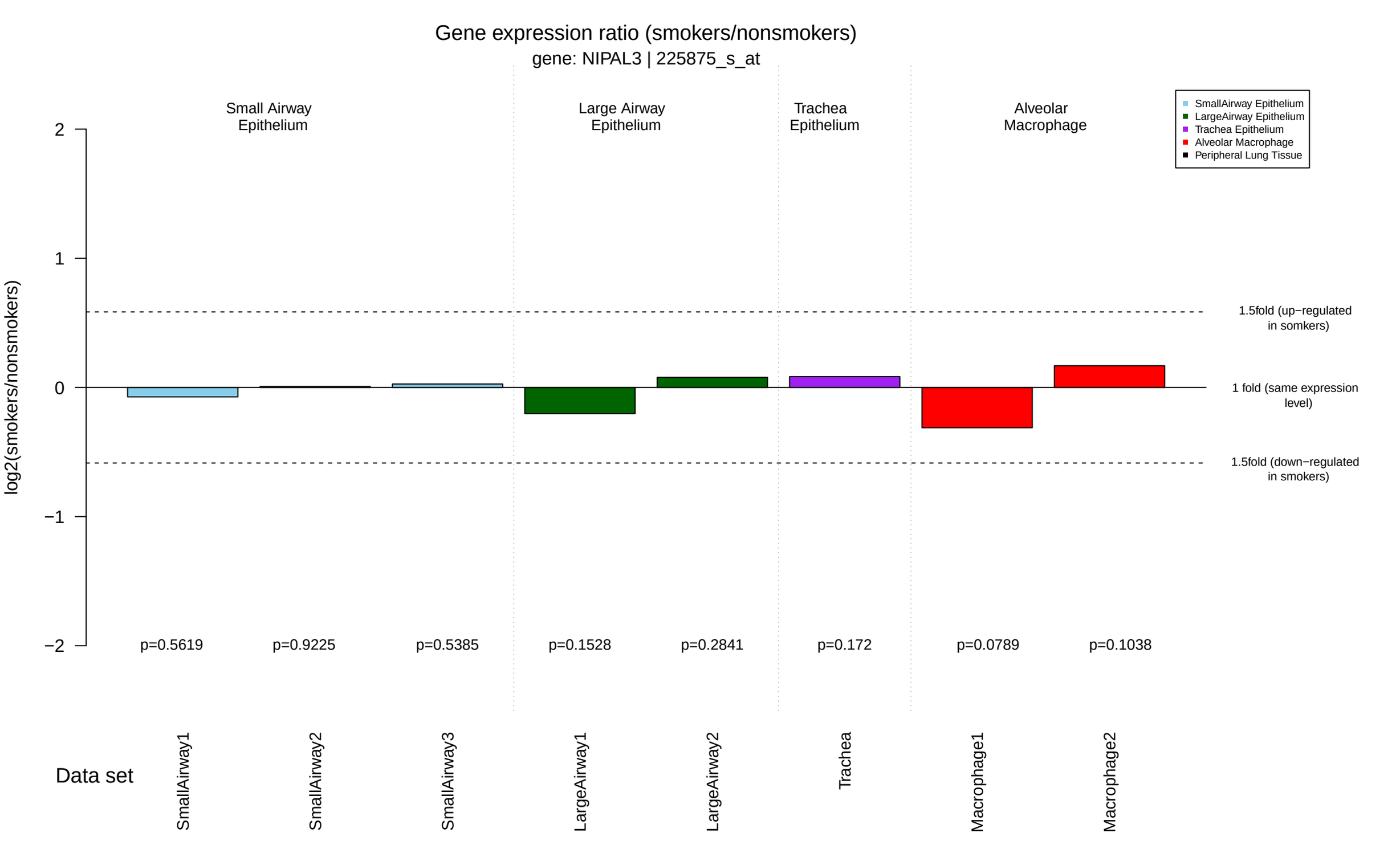
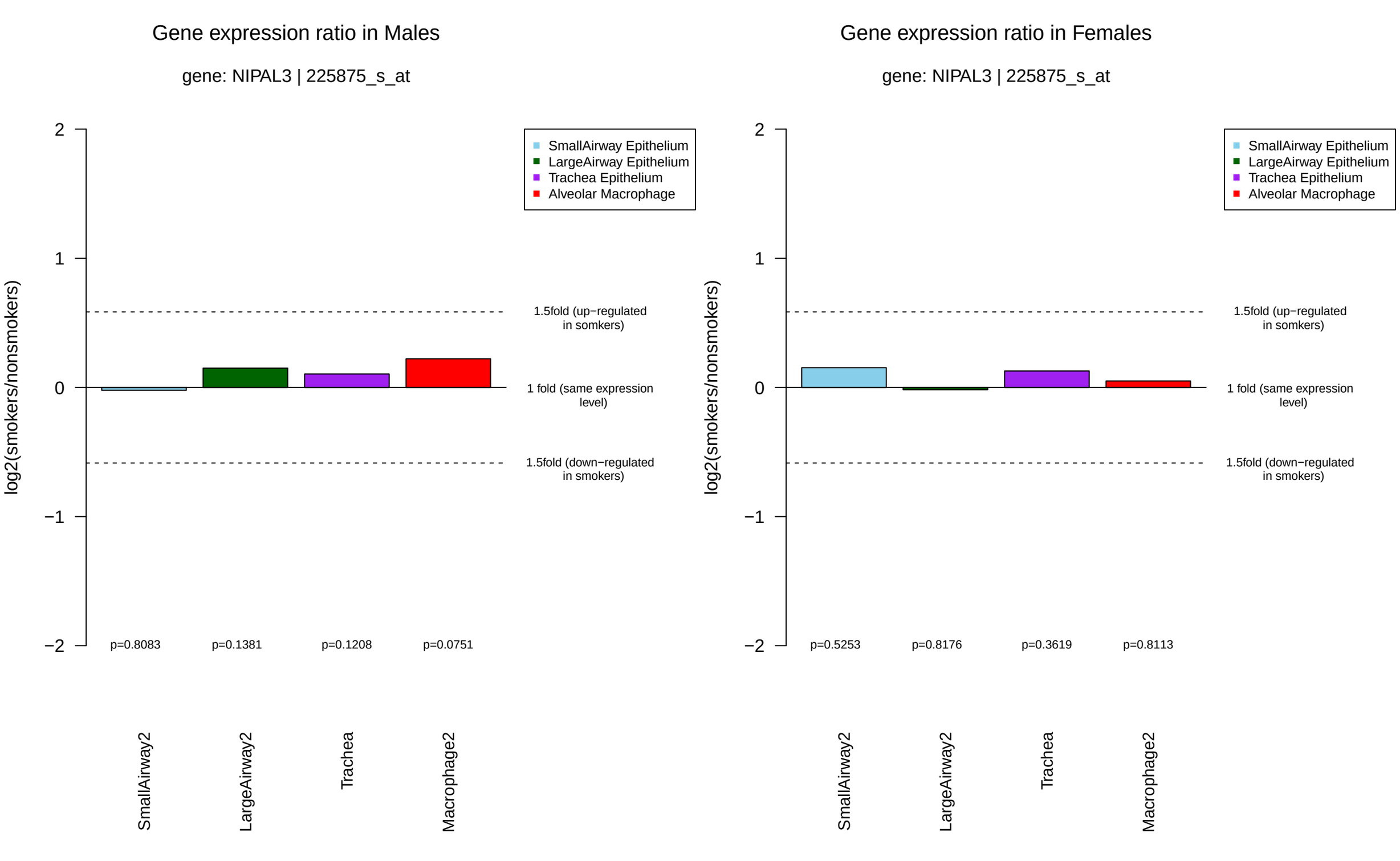
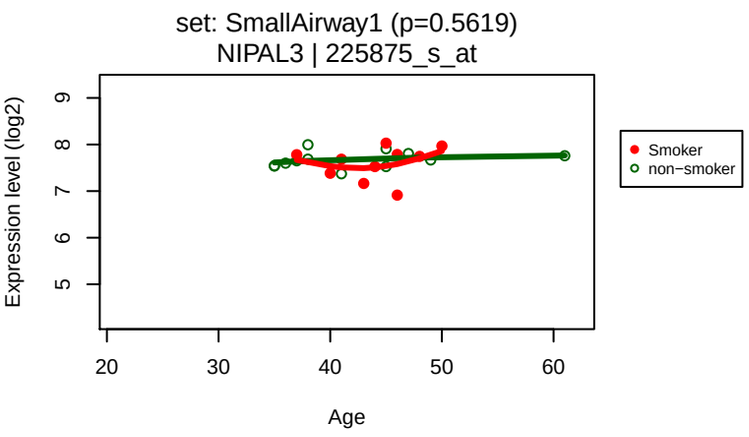
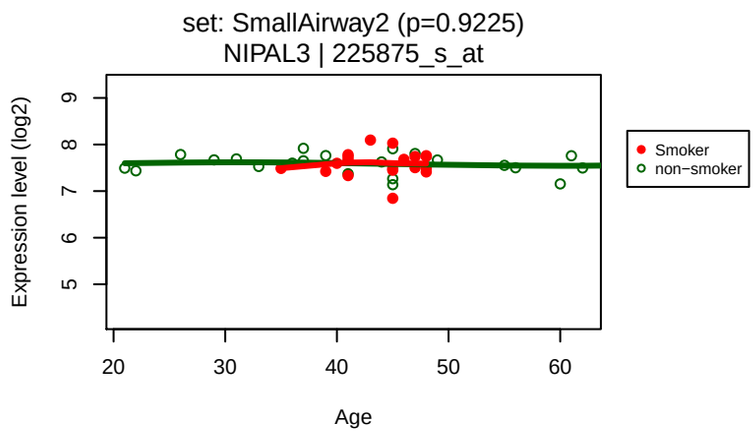
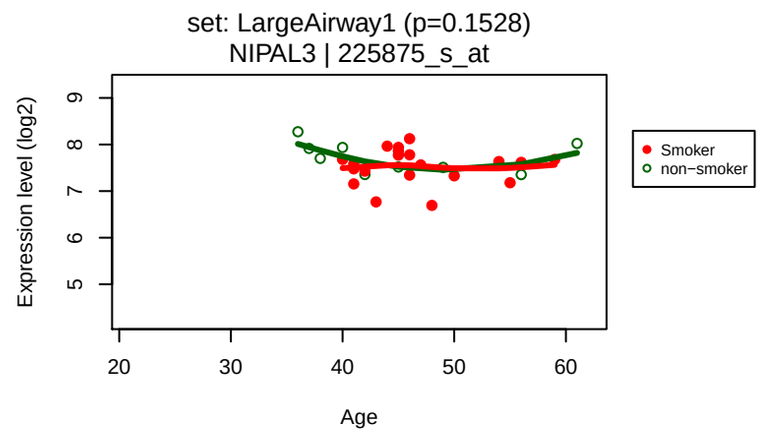

Gene expression of smokers and nonsmokers
gene: NIPAL3 | 225875_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5619
p=0.9225
p=0.5385
p=0.1528
p=0.2841
p=0.172
p=0.0789
p=0.1038
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

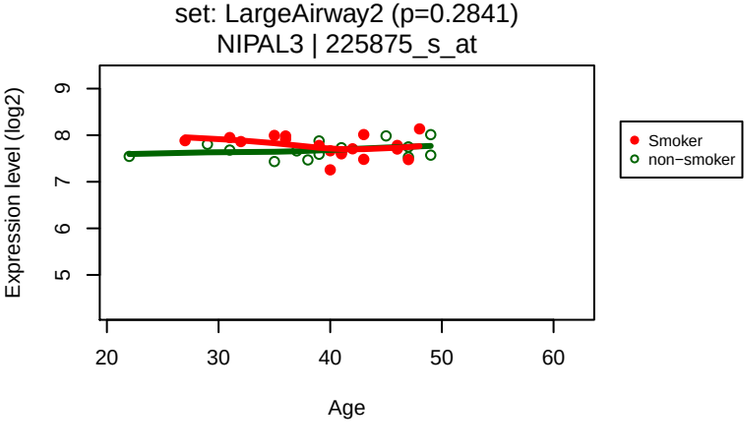

Gene expression ratio (smokers/nonsmokers)
gene: NIPAL3 | 225875_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5619
p=0.9225
p=0.5385
p=0.1528
p=0.2841
p=0.172
p=0.0789
p=0.1038
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

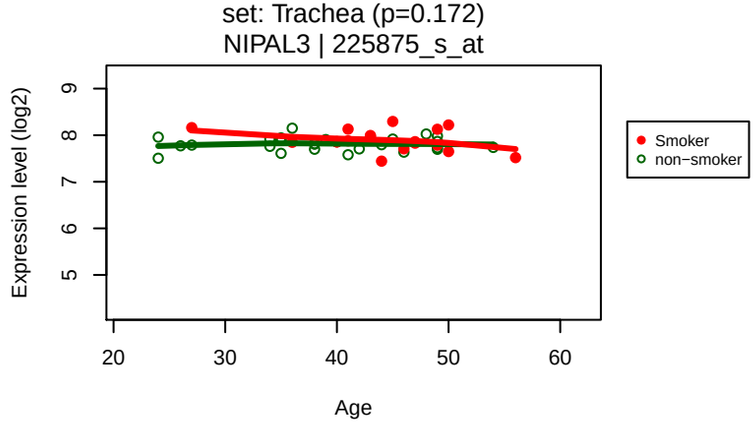

Gene expression ratio in Males
gene: NIPAL3 | 225875_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8083
p=0.1381
p=0.1208
p=0.0751
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NIPAL3 | 225875_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5253
p=0.8176
p=0.3619
p=0.8113
SmallAirway2
LargeAirway2
Trachea
Macrophage2

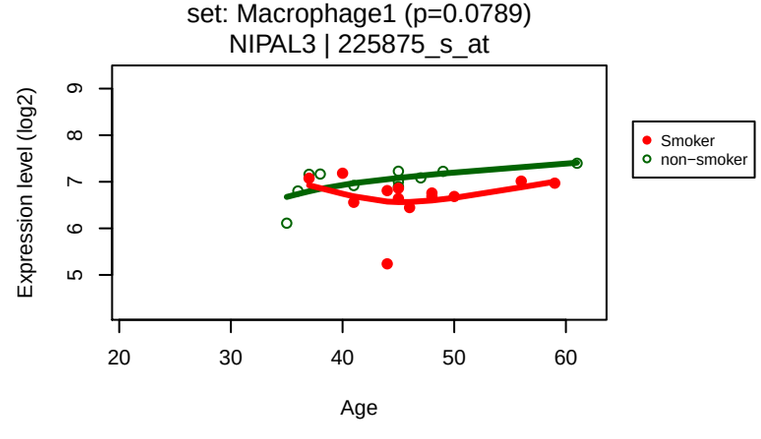

set: SmallAirway1 (p=0.5619)
NIPAL3 | 225875_s_at
Expression level (log2)
Age
Smoker
non−smoker

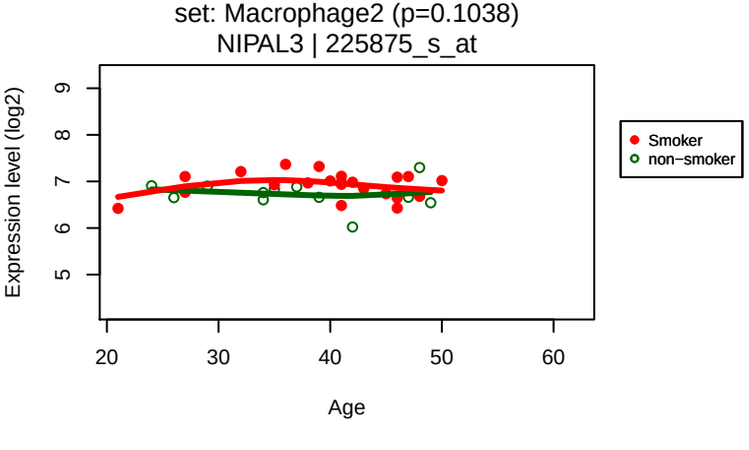

set: SmallAirway2 (p=0.9225)
NIPAL3 | 225875_s_at
Expression level (log2)
Age
Smoker
non−smoker

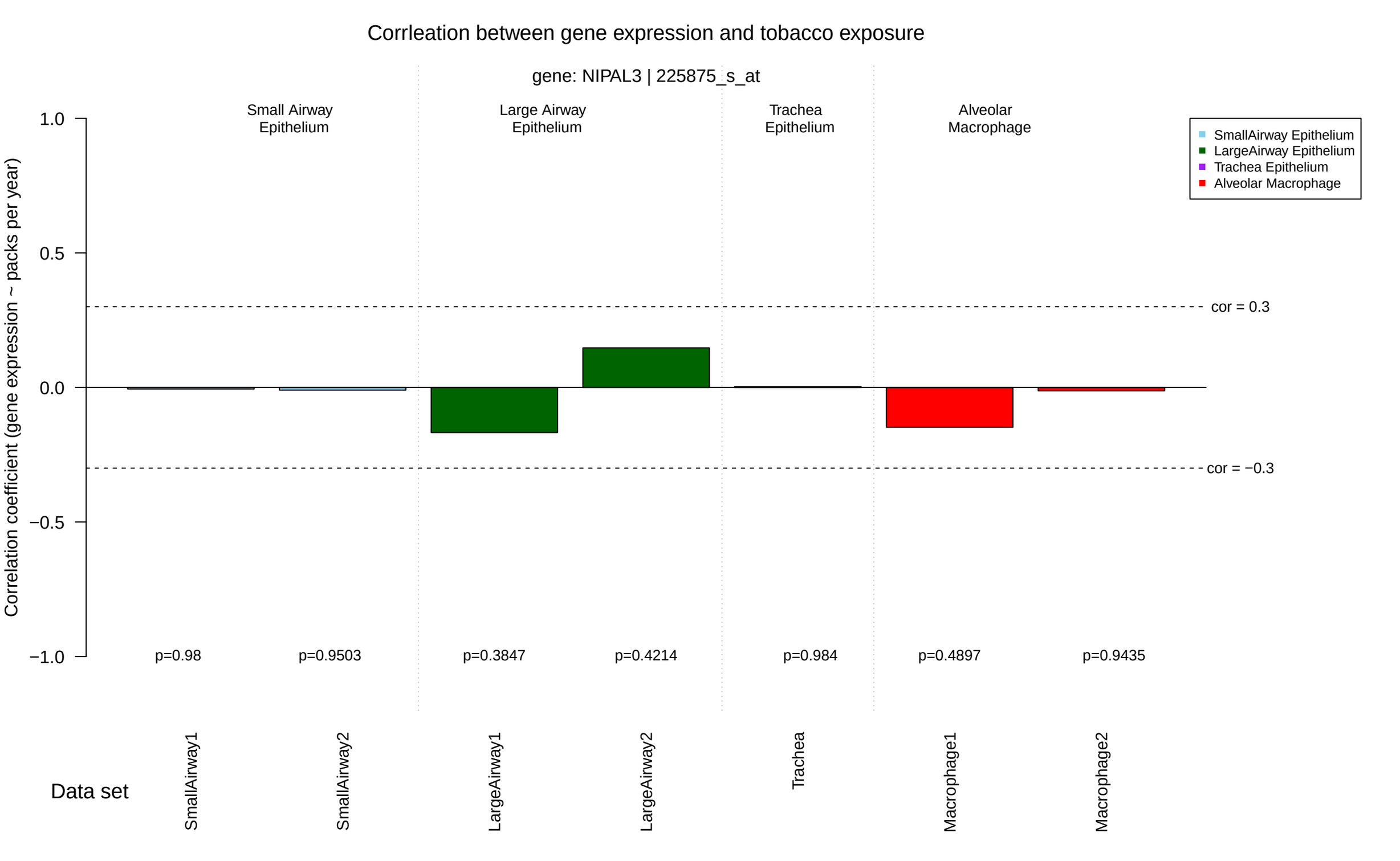

set: LargeAirway1 (p=0.1528)
NIPAL3 | 225875_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2841)
NIPAL3 | 225875_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.172)
NIPAL3 | 225875_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0789)
NIPAL3 | 225875_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1038)
NIPAL3 | 225875_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NIPAL3 | 225875_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.98
p=0.9503
p=0.3847
p=0.4214
p=0.984
p=0.4897
p=0.9435
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2